PRODUCT OVERVIEW · FOR DEVELOPERS AND HOSPITAL STAFF

# How a hospital works, and the system we will build for it

End-to-end workflow, paperwork, modules and architecture for a national HMS serving 1,000 to 10,000 hospitals.

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# What this deck covers

PART 1

## How a hospital is organised

Departments, roles and who does what

PART 2

## The patient journey and its paperwork

OPD, IPD and emergency, form by form

PART 3

## Money, supplies and people

Billing, insurance, pharmacy, store, HR

PART 4

## What we build and how

Modules, architecture, tenancy, standards, roadmap

# Every hospital has three kinds of departments

## Clinical

Treat the patient

- OPD clinics by speciality
- Emergency / casualty
- Wards, ICU, NICU
- Operation theatre, cath lab
- Nursing

## Clinical support

Help doctors decide and treat

- Laboratory, pathology
- Radiology: X-ray, CT, MRI
- Pharmacy
- Blood bank, CSSD
- Dietary, physiotherapy

## Administration

Keep the hospital running

- Front office, registration
- Billing, insurance / TPA
- Store and purchase
- HR, payroll, roster
- Medical records (MRD)

# The patient journey, start to end

EVERY PATIENT

1 Arrival → 2 Registration → 3 Triage, token → 4 Consultation → 5 Lab, radiology → 6 Review, Rx

OPD · GOES HOME SAME DAY

**7 Pharmacy → 8 Billing → Home**

About 80–90% of visits end here

IPD · ADMITTED

**9 Admission → 10 Pre-auth → 11 Ward → 12 OT → 13 Discharge → 14 Follow-up**

**Emergency** patients skip the queue: triage and treatment first, registration and billing after the patient is stable.

OUTPATIENT (OPD)

# A visit in eight steps, and the paper behind each

| Step | Who | Paper form today | In the software |
|---|---|---|---|
| Arrival | Help desk | Enquiry slip | Online booking, kiosk check-in |
| Registration | Front desk | Registration form, UHID card | Patient master, duplicate check, QR card |
| Triage | Nurse | Token and vitals slip | Live queue, SMS when turn is near |
| Consultation | Doctor | OPD case sheet | EMR templates, ICD-10 diagnosis |
| Investigations | Lab, radiology | Requisition, sample label | Orders, barcodes, analyzer results |
| Review | Doctor | Prescription | e-Prescription, interaction alerts |
| Pharmacy | Pharmacist | Issue slip, drug register | Auto-pick from e-Rx, live stock |
| Billing | Cashier | OPD bill, receipt | Charges auto-posted, UPI, GST invoice |

INPATIENT (IPD)

# From admission to discharge

| Step | Who | Paper form today | In the software |
|---|---|---|---|
| Admission | Admission desk | Admission form, consents | IP number, bed board, e-consent |
| Pre-authorisation | TPA desk | Cashless request form | Pre-filled from EMR, status tracking |
| Ward care | Nurse, doctor | Nursing chart, MAR, round notes | Bedside charting, dose alerts |
| Indents | Nurse | Pharmacy and store indent | Ward indents, auto-charged to bill |
| Procedure, OT | Surgeon, OT team | OT booking, safety checklist | OT calendar, implant scan |
| Transfer | Ward staff | Transfer note | Bed-to-bed move, ICU tariff switch |
| Discharge | Doctor, billing | Discharge summary, final bill | Auto summary, insurer and patient split |
| Records | MRD | Paper file in records room | Online record, shareable by health ID |

EMERGENCY

# Treat first, paperwork later

1

### Triage

Colour code: red, yellow, green. Quick name and phone only.

2

### Stabilise

Orders and drugs given before any bill exists.

3

### Legal (MLC)

Accidents, assault, poisoning need a medico-legal case record.

4

### Decide

Admit, transfer, or discharge home. Register fully now.

**For developers:** allow an "unknown patient" record that merges into the real UHID later, and never block treatment on a pending payment.

# Each paper form becomes a screen and a table

**Registration form**
→ patient, address, payer, consent

**OPD case sheet**
→ visit, vitals, notes, diagnosis

**Prescription**
→ medication order, dose, duration

**Lab requisition**
→ order, sample, result, report

**Admission form**
→ admission, bed, attendant

**Nursing chart / MAR**
→ observation, dose given, note

**Consent forms**
→ signed document, witness, time

**Bill and receipt**
→ charge lines, payment, refund

**Discharge summary**
→ summary document, follow-up

Tip: before you design any screen, collect the real printed form from 3–5 hospitals and map every field.

REVENUE CYCLE

# How money flows through a hospital

**Tariff master**
Prices per service, ward class, payer, package

→

**Charges post**
Each test, drug, bed-day adds a line

→

**Bill**
Interim and final, discounts, GST

→

**Payer split**
Patient, insurer, govt scheme, company

→

**Settlement**
Claims, deductions, accounts

### Cash / UPI

Advance deposit at admission, top-ups, refund at discharge

### Insurance / TPA

Pre-auth, enhancement, final claim with documents

### Government schemes

Fixed package rates, portal submission

# Behind the care: supplies and people

### PHARMACY AND STORE

1. Ward raises an indent
2. Store checks stock, raises purchase request
3. Purchase order to supplier
4. Goods receipt (GRN) with batch and expiry
5. Issue to ward or pharmacy counter
6. Charge to patient, reorder alert

### PEOPLE

1. Staff master with licences and expiry
2. Duty roster: doctors, nurses, shifts
3. Doctor schedules feed appointment slots
4. Attendance and leave
5. Payroll and doctor revenue share
6. Role and department access rights

# Modules, grouped by when to build them

### CORE · EVERY HOSPITAL

- Patient registration, UHID
- Appointments and queue
- OPD and EMR
- Pharmacy
- Billing and receipts
- Users, roles, settings
- Basic reports

### GROWTH · MEDIUM HOSPITALS

- IPD, admission, beds
- Nursing and MAR
- Laboratory (LIS)
- Radiology (RIS)
- Insurance / TPA
- Inventory and purchase
- Patient mobile app

### ADVANCED · LARGE AND CHAINS

- Emergency and MLC
- OT and cath lab
- ICU charting
- Blood bank, CSSD, diet
- HR, roster, payroll
- Multi-branch, PACS
- Analytics, national ID

# System architecture in six layers

| | |
|---|---|
| **1 Users** | Patients, front desk, doctors, nurses, lab, pharmacy, admin, platform team |
| **2 Channels** | Staff web app, patient app, doctor tablet app, kiosk, super-admin console |
| **3 Edge and access** | CDN and firewall · API gateway resolves the hospital · SSO with role-based access |
| **4 Services** | Registry, OPD, IPD, ER, lab, radiology, pharmacy, billing, insurance, HR, reports |
| **Event bus** | order.placed · result.ready · patient.admitted · bill.finalised |
| **5 Data** | PostgreSQL, Redis, file storage, search, warehouse, audit log |
| **6 Outside systems** | National health ID, insurers, lab analyzers, PACS, payments, SMS / WhatsApp |

1,000 TO 10,000 HOSPITALS

# One codebase, three ways to store a hospital's data

SMALL · clinics

## Shared database

tenant_id on every row, enforced by row-level security. Cheapest; onboard in a day.

MEDIUM · 50–300 beds

## Schema per hospital

Own backups and restore, shared cluster. Supports branches.

LARGE · 300+ beds

## Dedicated database

Own replicas and reporting. On-premise option for those who need it.

Every request carries **hospital, branch and role**. No query ever crosses hospitals.

# Standards and rules the system must follow

### Data exchange

FHIR R4 for records, HL7 v2 and ASTM for lab machines, DICOM for images

### Clinical codes

ICD-10 diagnoses, LOINC lab tests, SNOMED CT terms, a national drug list

### Privacy and security

Consent, encryption, audit of every view, data kept in-country (India: DPDP Act 2023)

### National programmes

Health ID linking and accreditation needs (India: ABDM, NABH)

Indian examples shown. Confirm the current rules with a healthcare lawyer for your target country before launch.

# A phased rollout

PHASE 0

## Learn

Visit 3–5 hospitals, collect forms, shadow front desk and wards

PHASE 1

## Core

Registration, OPD, pharmacy, billing; pilot in 2–3 clinics

PHASE 2

## Growth

IPD, lab, insurance, inventory, patient app; onboard first 100

PHASE 3

## Scale

ER, OT, HR, PACS, national ID, analytics; self-serve onboarding

Durations depend on team size; set dates after Phase 0.

# Next steps

1. Watch the patient-journey walkthrough together with hospital staff
2. Collect real forms from pilot hospitals and map every field
3. Agree the core modules and the tenancy model
4. Design the patient, visit and billing data model first